What Could the Agent See at 19:05?

Generating Temporal Enterprise Scenarios from Real Research
and Replaying Them to Evaluate Agents

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An answer is correct only relative to **what data existed** and **who could see it** at the moment it was asked.

We **generate** a believable, time-evolving enterprise world from real research and **replay** it at any instant, turning one episode into dozens of point-in-time, per-person, byte-for-byte reproducible evals.

When the evaluator can't rewind

Offline evals grade against one frozen **snapshot** — effectively the **end of the story**.
But real work is a story unfolding over hours across many apps.

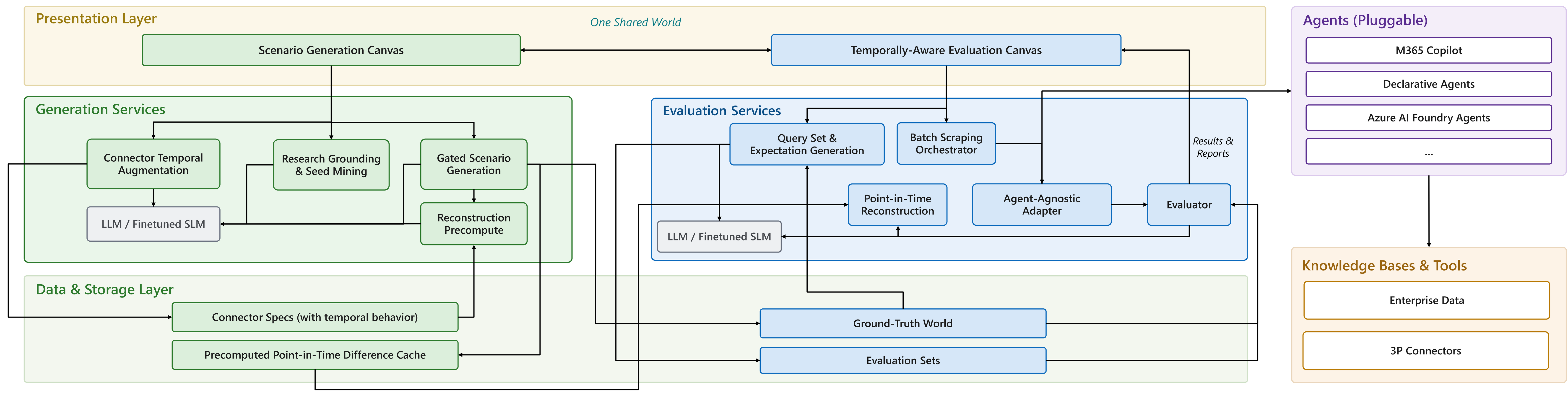
- **Only the final situation is evaluable.** The rich middle — where most questions are asked — is unreachable.
- **The future leaks from inside records.** Dropping rows "newer than t" misses state buried within a record (later comments, a resolution).
- **Hand-built ground truth doesn't scale.** One believable multi-app episode takes a skilled person days.
- **Always-on agents can't be evaluated at all.** Nothing can place a react-to-change agent just after the change.

A concrete episode: A payments service degrades; over two hours an incident is opened, acknowledged, triaged, fixed by a merged PR, and resolved with a root cause.
At **19:05** (mid-triage) Rohan asks: *"what's the status of my incident, and what's on the timeline so far?"*

The same incident record — the truth at 19:05 vs the end-state snapshot the agent is graded against.

Field	Truth @ 19:05	End-state snapshot
status	acknowledged	resolved
timeline	3 entries	10 entries
root cause	(none yet)	full RCA text

System architecture *— two cooperating flows over one shared world*



SCENARIO GENERATION

Grounded in reality · Temporal by design
Built for evaluation

From real research → Believable, temporally-evolving worlds

OUTPUT Scenario package
One self-contained, all-knowing ground truth

1. **Inputs (Real World)**
   - User interviews
   - Customer DSATs
   - Support tickets
   - Feature requests
   - Incident reports
   - PM specs
2. **Seed Mining (Recurring Themes)**
   - Seed A
   - Seed B
   - Seed C

   Mine patterns across docs → concrete seeds
3. **Personas**
   - Rohan — on-call eng
   - Vikram — manager
   - Priya — PM

   roles, goals & access (visibility)
4. **Narrative Arc (Beats)**
   1. Incident triggered
   2. Acknowledged
   3. Config-change triage
   4. Fix opened & merged
   5. Resolved + RCA

   cause → effect spine
5. **Timeline (Events over time)**
   - 18:42 trigger · 19:05 triage · 20:38 merge · 20:55 resolve
   - $(T_1, E_1) \dots (T_n, E_n)$

   long-horizon episode
6. **Artifacts (package)**
   - Email
   - GitHub Issues/PRs
   - Teams
   - Confluence Docs
   - Jira Tickets
   - PagerDuty Alerts

TEMPORALLY-AWARE EVALUATION

Evaluate agents the right way — at the right time, with the right context

Timeline
Long-Horizon Episode

T_1	T_2	T_3	T_4	T_5	T_6
Alert fires	Teams triage	Jira ticket	Email update	GitHub fix PR	Postmortem

At any query time T_i:
Only events **up to T_i** are visible — to the agent *and* to the grader.

1. **Scenario Package**
   - Email · Teams · Jira · GitHub

   one all-knowing ground-truth world
2. **Eval-Set Generation**
   - Smart, scoped queries anchored to a **persona** & **timestamp**.
   - **Query set (TSV)**
   - **+ Checklist Assertions**
3. **Scoped Response Generation ★**
   - `Query (q) @ Time Instant (T₃)`
   - T1 · T2 · T3 · T4
   - Agent sees only events $\leq T_3$ across all sources (including connectors)
   - **Agent response r_3**
4. **Scoped Assertion Eval ★**
   - ✓ Status = acknowledged
   - ✗ Mentions merged fix
   - ✓ Timeline: 3 entries
   - ✓ No root cause yet

   (each assertion graded with reasoning)
5. **Batch Scorecard**
   - 78% **Overall** Pass Rate
   - *by dimension · per-query drill-down*

★ *core technical novelties — temporal scoping of both the world the agent sees and the grader's expectations*

Reconstruction: the math

$$\rho(a,\, t) \;=\; \Lambda(D(a,\, t),\, t)$$

Two-stage rebuild of record a as of t —
D deterministic & cheap, Λ interpretive & rare.

$$\delta(a,\, t_i) \;=\; \rho(a,\, t_i) \ominus a^*$$

Queryable instants are finite — *precompute once; store only the tiny difference from final state a*.*

$$\Sigma(t, P) = \{\, a^* \oplus \delta(a,t) : \tau(a) \leq t,\; P \in \mathrm{acc}(a) \,\}$$

Replay is a pure-data lookup + merge, access-scoped —
no model in the path ⇒ reruns are byte-identical.

Reconstruction in action

Incident connector's temporal schema
Deterministic (D) vs Semantic (Λ) fields, rebuilt at 19:05.

Field	Kind	Rebuilt @ 19:05
`timestamp`	D	18:42 (kept)
`timeline[]`	D	10 → 3 entries
`resolved_at`	D	20:55 → null
`root_cause`	D	RCA → null
`status`	Λ	resolved → acknowledged
`description`	Λ	reworded to 19:05

Early payoff on enterprise agents

- One episode → dozens of point-in-time, per-person evals.
- Onboarding a new app: **days → minutes.**
- Batch runs are **byte-for-byte reproducible.**

Design lessons & what it beats

- **Push nondeterminism out of the measured path.** Model calls run before grading — reproducibility for free.
- **Generate in an order that forces coherence.** persona → arc → timeline → artifact bakes in cause-and-effect.
- **Separate "cheap & exact" from "careful & rare."** Most rebuilds never touch a model; a small remainder needs one.

Our system vs how agents are evaluated today.

Capability	Static	Simulated	Ours
Evaluate at any moment	—	—	✓
Correct nested record state	—	—	✓
Reproducible model-derived state	—	—	✓
Machine-generated expectations	man.	man.	✓
Any agent via one adapter	—	—	✓

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